

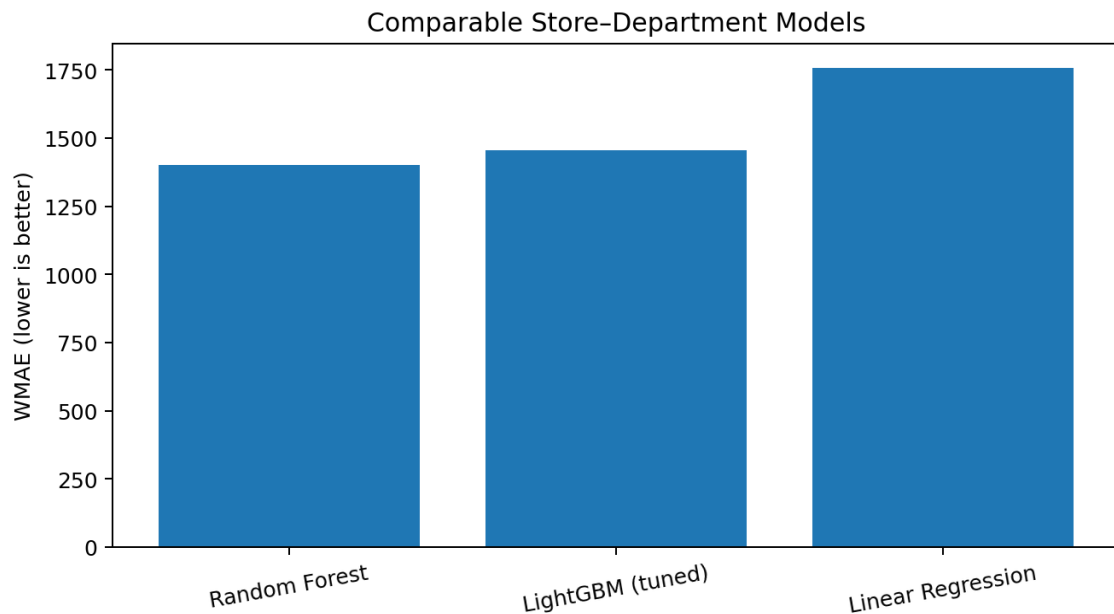
Walmart Store Sales — Model Comparison Report

Best Model Selection, Evaluation, Robustness & Explainability

Final selection: Random Forest — selected programmatically by minimum WMAE.

1. Executive Summary

The notebook evaluates three directly comparable Store+Department models: Linear Regression, Random Forest and tuned LightGBM. WMAE is the primary selection metric because holiday weeks receive 5× weight.



Random Forest has the lowest WMAE at 1,400.53. LightGBM has a marginally lower RMSE and marginally higher R^2 , but its WMAE is higher. Therefore Random Forest is the correct final selection under the notebook's stated business objective.

2. Comparable Model Performance

Model	MAE	RMSE	WMAE	MAPE	R ²
Random Forest	1,371.55	2,875.41	1,400.53	59.40%	0.98553
LightGBM (tuned)	1,407.72	2,866.75	1,456.44	170.93%	0.98562
Linear Regression	1,731.64	3,323.84	1,757.65	423.51%	0.98067

3. Best Model Reasoning

Accuracy: Random Forest has the best MAE and WMAE among comparable models. Its WMAE is about 3.8% lower than tuned LightGBM and about 20.3% lower than Linear Regression.

Business alignment: WMAE weights holiday errors five times more heavily, matching the higher inventory-planning cost of peak-period forecast misses.

Nonlinear signal: demand depends on interactions between store/dept identity, recent sales, annual seasonality, promotions, holidays and macro variables. Random Forest captures these interactions without requiring a linear form.

Operational trade-off: Linear Regression is more interpretable and faster, while LightGBM is a strong challenger. Random Forest provides the best WMAE while remaining practical for weekly batch inference, and SHAP supplies the needed explanation layer.

4. Validation Strategy

The primary hold-out is chronological: training through 2012-07-20 and validation from 2012-07-27. Random splitting is inappropriate because it can let future patterns influence training.

LightGBM tuning uses five expanding-window TimeSeriesSplit folds. Fold MAEs were 1,563.29; 1,469.34; 2,930.84; 1,776.86; and 1,520.72, with mean 1,852.21 and standard deviation 549.37.

5. Prophet and LSTM

Prophet and LSTM are included as exploratory models at aggregate total-Walmart level. Their metrics are intentionally kept separate because they do not predict the same Store+Department target granularity.

Model	MAE	RMSE	WMAE	MAPE	R ²
Prophet (aggregate)	544,562.83	760,184.21	538,914.47	2.14%	0.40645
LSTM (aggregate)	764,688.37	973,215.19	773,183.88	3.01%	0.02717

6. Robustness Testing

The winning model is evaluated across the assignment's required segments.

Holiday vs non-holiday

Group	Count	MAE	WMAE
Non-Holiday	18,641	1,361.52	1,361.52
Holiday	1,431	1,502.14	1,502.14

Store type

Group	Count	MAE	WMAE
A	9,837	1,716.26	1,739.93
B	10,235	1,040.24	1,073.77

Markdown level

Group	Count	MAE	WMAE
Low	64	708.88	708.88
Medium	2,277	857.30	829.05
High	17,731	1,439.98	1,471.81

The most important robustness signal is the store-type gap: Type A WMAE is 62.0% higher than Type B. High-markdown observations also have materially higher error.

7. Residual Analysis

Residuals are visualized versus predictions, as a distribution, and over time. Store-level and department-level error rankings are also reported. Store 14 has the highest store-level MAE (2,860.24); Department 38 has the highest department-level MAE (6,221.19). The notebook visually inspects residual distribution but does not implement a formal normality test.

8. Explainability

Rank	Feature	Mean SHAP
1	Sales_lag_52	9,047.75
2	Sales_lag_1	7,904.74
3	Sales_roll_mean_4	799.21
4	Sales_roll_mean_8	307.44
5	Sales_roll_mean_12	154.89
6	Sales_lag_2	152.85
7	Sales_lag_4	123.40
8	Unemployment	91.61

Rank	Feature	Mean SHAP
9	Days_to_Holiday	84.29
10	MarkDown3	80.32

The top SHAP features show strong yearly seasonality, short-term demand momentum and recent rolling demand. Days_to_Holiday and markdown activity provide additional business-relevant explanatory signals. Three local SHAP examples and a LIME cross-check are also generated.

9. Trade-off Analysis

Dimension	Linear Regression	Random Forest	Tuned LightGBM
Accuracy	Lowest	Best WMAE	Close; best RMSE/R ²
Interpretability	High	Medium + SHAP	Medium + SHAP
Training	Very fast	Moderate	Moderate + tuning
Nonlinear signal	Limited	Strong	Strong
Deployment	Very simple	Simple batch	Simple–moderate
Selected	No	Yes	No

10. Business Context

This is a weekly inventory-planning problem rather than a real-time transactional prediction problem. Under-forecasting holiday demand can cause stockouts; over-forecasting can increase excess inventory and markdown exposure. Selecting by WMAE aligns the model with this cost structure.

11. Limitations & Improvements

- Holiday and Type A errors are higher than their comparison groups.
- Recursive forecasting can propagate prediction errors into future lag features.
- The current cold-start fallback fills unavailable lag/rolling features with zero; a production system should use hierarchical or store-type/department fallback estimates.
- Future improvements: holiday-specific modeling, hierarchical forecasting, richer external signals, ensembling, and automated drift monitoring.

12. Final Recommendation

Use Random Forest as the final baseline forecasting model. It is the strongest model under the primary WMAE objective while maintaining strong overall predictive performance. Retain tuned LightGBM as the principal challenger for future tuning and ensemble experiments.